

E: Efficient Encoding

Source file name: `encoding.c`, `encoding.cpp`, `encoding.java`, or `encoding.py`

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It is the year 3057. The Interplanetary Compression Priority Code (ICPC) is tasked with maintaining communications among human colonies scattered across the galaxy. To survive in a hostile environment with severely limited bandwidth, every message must be compressed as efficiently as possible before transmission.

Each message is a multiset of interplanetary symbols. The ICPC compresses messages written with an alphabet of n symbols, assigning to each symbol a codeword constructed using a binary prefix-free code (i.e., no codeword is a prefix of another). As usual, the ICPC relies on optimal coding strategies that minimize the total number of bits required—just as the legendary compression schemes of the 21st century once did.

However, the number of occurrences of each symbol is not constant over time (the ICPC actually works with their expected values). They fluctuate due to solar storms, the rotation of the quantum core, and other cosmic phenomena that only astrophysicists pretend to understand. To cope with this variability, between time 0 and a fixed time upper bound T , the ICPC models the expected value of the number of occurrences of each symbol as a piecewise-linear function specified by three points with coordinates in the non-negative real numbers (i.e., in $\mathbb{R}_{\geq 0}$): $(0, a_i)$, (m_i, b_i) , and (T, c_i) , with $0 < m_i < T$, and a_i, b_i , and c_i integers. Between these points, the expected value for the number of occurrences of i th symbol, $f_i(t)$, varies linearly over real values. At any time $t \in [0, T]$, the expected value of the number of occurrences of symbol i is:

$$f_i(t) = \begin{cases} a_i + \frac{b_i - a_i}{m_i} \cdot t, & 0 \leq t \leq m_i, \\ b_i + \frac{c_i - b_i}{T - m_i} \cdot (t - m_i), & m_i \leq t \leq T. \end{cases}$$

To choose the best encoding time, the ICPC considers, for each $t \in [0, T]$, the optimal binary prefix-free code for the frequencies $\{f_i(t)\}_{i=1}^n$ and evaluates its total cost

$$C(t) = \sum_{i=1}^n f_i(t) \ell_i(t),$$

where $\ell_i(t)$ is the codeword length assigned to symbol i by an optimal code at time t .

Your task is to help the ICPC choose the ideal encoding time: find the minimum time $t \in [0, T]$ that minimizes the total encoding cost $C(t)$.

Input

The first line contains the number of test cases K ($1 \leq K \leq 10^3$). For each test case, the first line contains two integers n and T ($1 \leq n \leq 100$ and $1 < T \leq 10^6$), denoting the number of symbols and the maximum time value, respectively. Each of the next n lines contains four integers a_i, m_i, b_i , and c_i ($0 < m_i < T$ and $0 \leq a_i, b_i, c_i \leq 10^6$), describing the frequency function $f_i(t)$.

The input must be read from standard input.

Output

For each test case, output two blank-separated numbers on one line: the smallest integer time $t \in [0, T]$ that minimizes $C(t)$, followed by the minimum cost $C(t)$ rounded to three digits after the decimal point.

The output must be written to standard output.

Sample Input	Sample Output
2	0 14.000
3 10	1 1.000
2 5 10 4	
4 6 9 3	
3 8 8 7	
1 2	
2 1 1 2	